

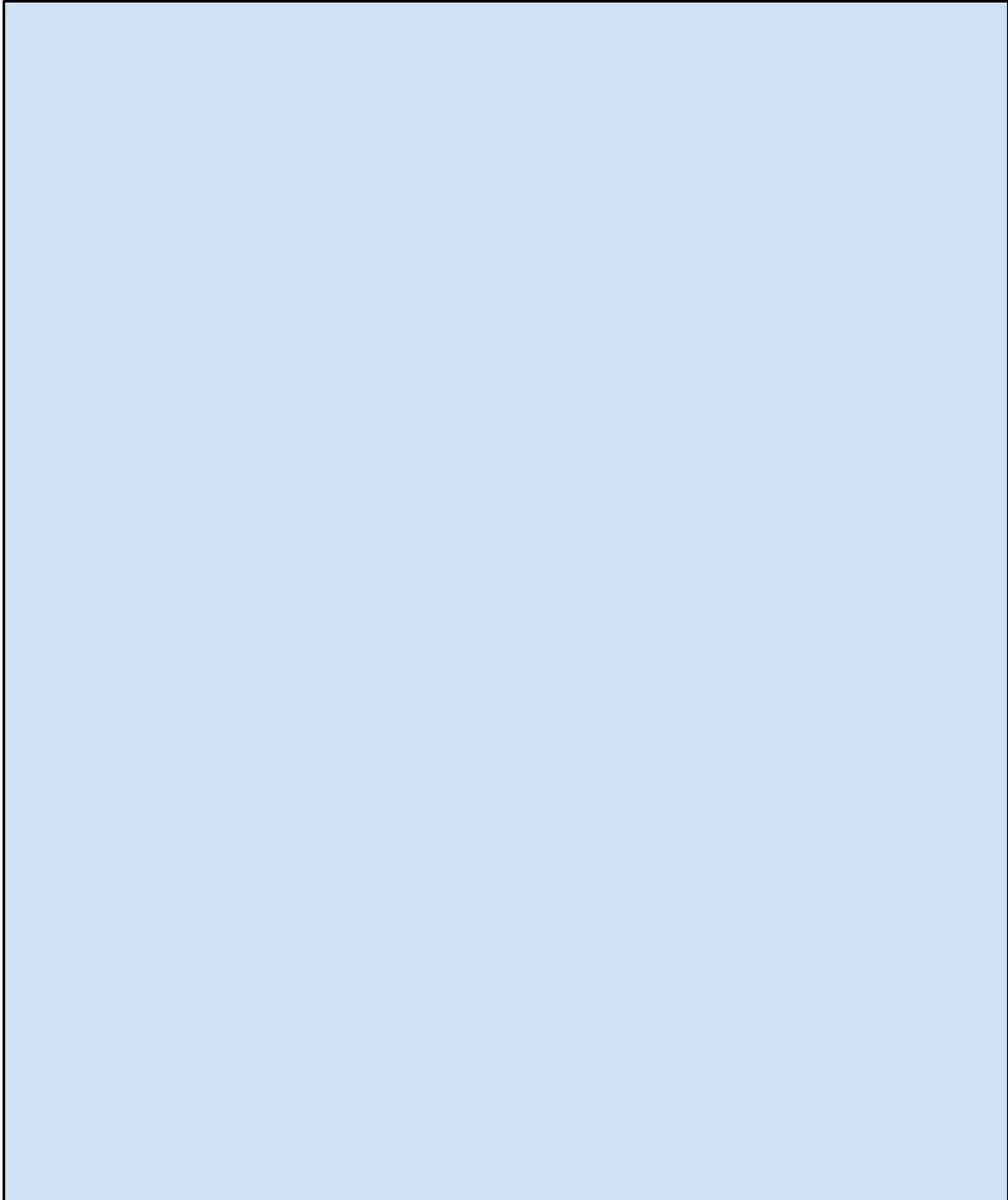


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1.0 Design Thinking Process

Step 1: Empathize with potential customers

To understand customers' motivations and behaviors, we have conducted interviews with 10 participants aged 20–50, selected through purposive sampling to represent different gifting motivations, including romantic, friendship, family-oriented, self-gifting, and occasion-driven (*Appendix A to C*). We asked questions about their gift-giving habits, their desire for personalized and meaningful gifts, and their preferences in jewelry.

Step 2: Define users' needs and problems

Based on the interview findings, we identified several customer pain points and unmet needs. These insights can be grouped into three key areas:

a. Personalized and Meaningful Gifts Matters Deeply

From our interviews, participants highly value gifts that feel personal and meaningful. Whether giving or receiving, they prefer the gifts that reflect the relationship, shared memories, or the recipient's personality, creating a genuine emotional connection. For them, thought and sentiment matter far more than price or appearance. This deep desire for emotional resonance drives people to seek gifts that are memorable, thoughtful, and meaningful.

b. Difficulty Finding Unique Gifts

Despite this strong desire for personalized gifts, interviewees pointed out that they often struggle to find options that truly meet their needs. As most “personalized” options on the market still feel impersonal. Many offerings are mass-produced and only allow minimal customization, like adding initials or selecting a color. As a result, these gifts often come across as generic and fail to convey the emotional authenticity and sentiment they want to express with the recipients.

c. Budget is a Major Barrier

Our interviews showed that many users find high-quality jewelry too expensive, making it difficult to afford for most gift-giving occasions. This creates a challenge: cheaper options often feel generic, while higher-end pieces exceed their budget. People want gifts that feel thoughtful and personalized, but that still fit within a reasonable price.

Step 3: Ideate Innovative Solutions

Then, moving into Step 3, Ideate, our team developed creative solutions based on the pain points we discovered. Inspired by a student pitch to Pandora based on *Appendix E*, we focused on enhancing traditional jewelry with modern technology to create deep, lasting personal value. Our core solution is NFC-Enabled Memory Charms as a bracelet jewelry called **ETÉRA**. The way it works is, each collectible is embedded with a Near Field Communication (NFC) chip. Using a companion app, the giver can link said charm to a private digital album to commemorate a specific memory, forever. The objective to transform jewelry into an interactive, living memory that achieved personalisation, unique while being affordable.

2.0 Market Validation

To validate the feasibility of **ETÉRA**, we prepared a 20-question survey and successfully collected responses from 101 participants seen in *Appendix D*. The survey results indicate strong market potential driven by participants' gifting behavior. Specifically, 77.3% of respondents reported they frequently buy gifts, and 78.2% said meaningful and personalization as the most important gift selection criteria. Furthermore, 88.1% gave 3 stars and above find it difficult to choose meaningful gifts, proving a clear demand for more emotionally expressive products.

Although 55.4% of respondents were unfamiliar with smart jewelry, 67.3% already wear general jewelry, suggesting limited local market exposure but still have potential for adoption. The emotional and technological features were highly appealing, with 96% found it interesting to gift jewelry that can store personal memories such as photos, videos or voice messages that can be unlocked through NFC scanning.

Regarding price, 99% of participants are willing to pay more for sentimental technology. With 52.5% considering RM151–RM300 reasonable for its features, components, and a premium gift product positioning. In terms of product type, 74.7% preferred the form of a bracelet, providing a clear direction for our first launch.

In conclusion, the market validation shows that NFC smart jewelry has high acceptance and strong demand potential, supported by current gifting trends, emotional value, and consumers' readiness to adopt technology-enhanced accessories. This confirms the practicality and attractiveness of introducing our product into the market.

3.0 Market Trends

In the **global market**, smart jewelry is experiencing strong growth, as fashion merges with wearable technology. Tech-enabled jewelry generated approximately USD 153.6 million in 2024 and is projected to reach around USD 430 million by 2030 (Grand View Research, n.d.b). The global jewelry market is growing steadily, projected from USD 366.8 billion in 2024 to USD 578.5 billion by 2033, providing a strong baseline of consumers familiar with jewelry purchases (Grand View Research, n.d.a).

In **Malaysia market trends**, the general jewelry market is growing steadily, whereas the smart jewelry market is still emerging.. The watches and jewelry segment is projected to generate USD 2.64 billion in 2025, while the overall jewelry market is expected to reach USD 1.76 billion, with a CAGR of 4.23% from 2025 to 2030 (Statista, n.d.).

Given these global and local trends, there is a **clear market opportunity for ETÉRA NFC bracelets**, which combine fashion, technology, and emotional personalization. The product aligns with global demand for stylish and functional wearables, while filling a local market gap for meaningful, technology-enhanced jewelry. Although potential challenges include local price sensitivity and limited awareness about NFC-enabled jewelry, rising demand for innovative, personalized, and giftable products creates a favorable environment for ETÉRA to establish a strong market presence.

4.0 Competitive Analysis

		la parure	Gemory™	totwoo CONNECT TO LOVE	<i>Etera</i>
Smart Technology (NFC)	✗	✗	✓	✓	✓
App Connectivity (iOS & Android)	✗	✗	✓	✓	✓
Multimedia Upload	✗	✗	✓	✓	✓
Personalisation Variety	Basic Engraving	Basic / Fingerprint / Portrait Illustration Engraving	Photo & Video	Audio, Photo & Video	Written Message, Audio, Photo & Video
Pre-Personalised Before Delivery (Gift-Ready Feature)	✓	✓	✗	✗	✓
Product Price (MYR)	RM 219 - RM 368	RM 55 - RM 129	RM 294	RM 250	Propose Price: RM 270

Personalised gifts are experiencing significant growth. However, gift buyers often encounter difficulty in finding one that brings about real emotions as opposed to the generic alternatives. Hence, to seize this opportunity, understanding where competitors fall short is key.

Analysis of local and global brands highlights distinct patterns. The strengths of local competitors like Yuniq and La Parure are affordability and convenience in pre-personalization, making their jewellery accessible and gift-ready. Yet their biggest weakness is that personalisation is restricted to simple engravings, which only have superficial meaning (La Parure, n.d.; YOUNIQ, n.d.). Conversely, global competitors like Gemory and ToTwoo demonstrate strengths in advanced NFC capabilities and multimedia upload, which allow more interactive and emotive gifting. However, their greatest weakness is inability to personalise prior to delivery, which brings inconveniences to long-distance gifting or surprise gifting (Gemory, n.d.; Totwoo, n.d.).

These gaps highlight clear opportunities for ETÉRA. By combining smart technology with a unique pre-upload before delivery system, ETÉRA enables givers to pre-load messages, voice notes, photos, and videos before shipping. To align with market expectations, we propose a competitive price of RM270, which fits in the RM150-RM300 price tolerance that 52.5% of survey respondents suggested. Nevertheless, ETÉRA must be aware of emerging competitive pressures. The main threats come from global competitors accelerating innovation and potential feature imitation, reinforcing the necessity of strong branding, continual app improvement, and innovation.

5.0 Marketing Strategy

ETÉRA will adopt a multi-channel marketing strategy aligning with user discovery habits to bridge the physical-digital gap.

Digital Marketing

Primarily, digital marketing strategy ensures visibility when potential customers actively hunt for meaningful gifts (Akashkumar *et al.*, 2025). Since 47.5% of survey respondents had discovered jewelry via online advertisements, we will be employing targeted Google Display Ads on relevant platforms like relationship milestone tracker apps to capture high-intent buyers. This will be complemented by SEO with keywords like “long-distance anniversary gifts” to secure the search traffic.

Social Media Marketing

Our survey also indicates 81.2% respondents discover products through social media, and 59.4% trust family and friends’ recommendations. Therefore, we will digitize word-of-mouth by initiating a #MemoriesOnTap campaign, encouraging customers to share their emotional experiences with ETÉRA on social media platforms like TikTok. By converting these authentic unboxing events into user-generated content, we make customers into trusted advocates, effectively scaling personal recommendations to drive sales.

Traditional Marketing

Despite digital shifts, 45.5% of our respondents still value visiting physical stores. Since smart jewelry is a new concept, physical interactions are a primary mechanism of trust-building (Invoke Media, 2025). Thus, we will be launching interactive pop-up kiosks in malls during peak seasons, where potential customers get to test the NFC feature themselves.

6.0 Global Resources Sourcing

ETÉRA has a global sourcing strategy to secure essential materials, components, and skills necessary for manufacturing and sustainable growth. The sourcing strategy combines worldwide product sourcing with local sustainable manufacture to achieve high quality, cost effectiveness, and ethical responsibility.

Country	Cost Level	Quality	Lead Time	Minimum Order Quantity	Sustainability Practices	Challenges
Netherlands	High	World-class NFC performance	3-6 weeks	High	Energy-efficient production	High cost and long production times
Switzerland	Very high	High quality	4-6 weeks	High	Excellent sustainability and compliance	Costly and long lead time
China	Low	Good	1-2 weeks	Very low	Improving green manufacturing	Quality

Comparison of NFC chips from different country

From sourcing NFC chips and components, ETÉRA compares suppliers from many countries. China provides the most effective choice with low costs, flexible minimum orders, and fast lead times suited for a startup. It is supported by Shenzhen's global dominance in semiconductor and wearable-tech production (Statista, 2024). Although Switzerland and Netherlands have strong sustainability and technological standards, their long lead times and premium cost decrease practicality for our budget. Overall, ETÉRA will select China because of its affordability, production flexibility, and stable supply chain.

For the jewelry base materials, ETÉRA sourced the stainless steel materials. China is a best choice due to its extensive pool of certified suppliers and cost-effective production capabilities. As the world's largest jewelry manufacturing hub, China offers a wide range of options, from metals to completed components, providing flexibility in order size and design. China's mix of affordable price, scalability, and extensive RJC-certified availability makes it perfect for businesses emphasizing both certification and efficiency (Responsible Jewellery Council, 2024).

For sustainable packaging, ETÉRA will choose eco-friendly packaging-biodegradable paper and recyclable boxes. It was sourced from Malaysian companies such as Packist Malaysia and EcoPrint to reduce environmental pollution. Final assembly and quality testing will be done in Malaysia to support circular economy principles and promote sustainable local manufacturing (Ellen MacArthur Foundation, 2024).

7.0 Alignment with Global Sustainability Development Goals

First, ETÉRA supports **SDG 9: Industry, Innovation and Infrastructure** by modernizing the traditional jewelry sector through advanced technologies. This seamless fusion of craftsmanship and secure digital infrastructure not only enhances the gifting experience but also demonstrates how traditional industries can evolve through technological innovation. ETÉRA's approach highlights its commitment to technological advancement and value creation through innovative digital solutions.

Besides, ETÉRA aligns closely with **SDG 12: Responsible Consumption and Production**. Unlike traditional mass-produced gifts, which are often generic and quickly discarded, ETÉRA's pieces can include personal messages, photos, or videos, creating strong emotional connections between giver and receiver. Items with emotional value are more likely to be treasured and kept longer, extending their lifespan and reducing disposable consumption (Kowalski & Yoon, 2022). By encouraging lasting and valuable gifts, ETÉRA helps break the cycle of disposable consumption and supports sustainable purchasing, directly contributing to SDG 12.

8.0 Resources Assumptions

Most of ETÉRA's resources will consist of the material required to design and produce our smart jewelry. The primary components include NFC chips and stainless steel. NFC chips will be sourced from China. It is because China is widely recognised as providing affordable, high-quality electronics manufacturing, and flexible order quantities suitable for startups (Statis, 2024). For materials on jewelry bases, we aim to source stainless steel from China's own RJC-certified suppliers, ensuring that our products meet responsible sourcing and sustainability standards (Responsible Jewelry Council, 2024).

ETÉRA plans to rent a studio as an office and store. Here, ETÉRA will assemble jewelry pieces, program the NFC chips, conduct quality checks, and prepare shipments for customers. In terms of manpower, ETÉRA will require trained jewelry craftsmen and technicians to assemble and finish each piece. ETÉRA also needs a marketing team to promote our products across social media and a finance team to manage expenditure. ETÉRA should partner with the logistics companies to distribute our products.

Lastly, ETÉRA would like to collaborate with Taylor's university's School of Computing Science for integrating new mobile app aligns with smart jewelry. This is to support innovation and improve product functionality while ensuring that our jewelry remains both technologically functional and aesthetically appealing.

9.0 Risk Analysis

ETÉRA's NFC-enabled sentimental jewelry faces several key risks. Market risks such as rapid technological advances means that NFC standards evolving from shifting consumer preferences for better digital privacy could reduce demand (Quach *et al.*, 2022). There are also operational risks like supply chain disruptions, potential app malfunctions or even data breaches that could corrupt the stored content in the app (Bai, Chen, and Zhang, 2025). Financial risk is also a possibility as this is a technology that involves a high R&D and customer acquisition costs due to targeting a niche market (Bielialov, 2022).

Mitigation strategies include investing in scalable, updatable cloud infrastructure with strong encryption and fostering a strong brand community to communicate this data security protocols can reduce market risk. Diversifying NFC chip suppliers and implementing rigorous quality assurance testing reduces operational vulnerabilities. Lastly, adopting a phased market expansion strategy can also help manage financial exposure while building brand recognition (Karanunamurthy, 2023; Asakdiyah and Oskar, 2024).

2000 words excluding tables, captions, headings, sub-heading and citations

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Appendix A

Interview Method and Questions

Name	Age	Motivation
Adeebah Ubaid	21 y/o	Friendship
Xin Nee	21y/o	Self-giver
John	23 y/o	Self-giver
Christine	25y/o	Friendship
Anson	26y/o	Romantic
Nurul Batrisyia	26 y/o	Occasion-driven
Jue Hang	31 y/o	Occasion
Zu Xian	50 y/o	Family-oriented
Junaidah Rahmad	52 y/o	Family-oriented
Hashim Ismail	53 y/o	Romantic

Interview Questions

1. Do you usually buy gifts for XXX often?
2. What kind of occasions and how often do you usually buy gifts for XXX?
3. How important is it for you that a gift feels personal or meaningful?
4. When you're choosing a gift for someone, or yourself, what tends to matter most to you (looks, practical, meaning etc)?
5. What makes you want to give a gift (obligation, appreciation, celebration, expression) ?
6. Can you recall a gift that felt especially meaningful and what made it so special?
7. How often do you buy jewelry as a gift?
8. How do you feel about receiving jewelry as a gift?
9. What are your thoughts on using technology to make gifts more interactive or personal?
10. Have you ever struggled to find a gift that feels truly unique or special?

Appendix B

Interview Findings (User Needs and Problem)

Key Findings From Interviews: User Needs & Problems

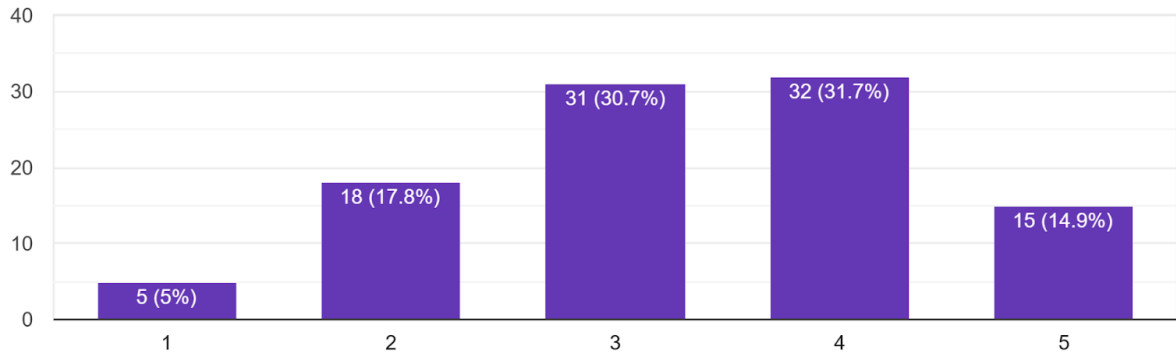
	User Need	Evidence
Need for personalisation and meaning over generic gifting	"Help me find a gift that feels uniquely tailored to the recipient's personality, interests, and our relationship."	Nurul, Adeebah & Anson explicitly look for gifts based on a person's wishlist or things they have said they wanted Adeebah and Nurul value sentimental value and gifts that remind them of the relationship. John and Adeebah appreciate technology that enables personalization (e.g., custom playlists, personalized items). Xin Nee and Christine prioritized emotional value and the gift-giving experience over the gift itself. The most meaningful gifts recalled were highly personal: a house, a handmade scrapbook, a box of 99 personal qualities, handwritten letter, handmade craft
The Practicality- Passion Balance	"I want my gift to be meaningful, but also useful and within a reasonable budget."	Junaidah, John & Zuxian prioritise "practical" gifts that can be used (e.g. pen, shoes, clothes) Adeebah & Christine specifically mentions staying "within my price range." Hashim is conflicted, citing both "practical" and "common romantic" gifts, indicating a struggle to merge meaning with convention. Joe Hang & Anson states he looks for items that are both unique and functional, avoiding overly decorative or purely symbolic gifts.
Difficulty and Effort of Finding "Unique" Gifts	"It's often stressful and challenging to find a gift that doesn't feel common or generic, especially for people I've known for a long time."	Adeebah & Christine states it's "hard to find for friend's I have been friends for a long time." John & Joe Hang explicitly says "yes" to struggling and dislikes "common things." He seeks limited or homemade items but finds custom options "too expensive." Nurul identifies that special gifts require more "effort" than common ones. Anson mentions he often spends extra time considering the recipient's personality and past comments to avoid generic gifts. Xin Nee notes that when the recipient "already has everything," the challenge of finding a unique gift increases.

Appendix D

Survey Results (101 Respondents)

How often do you buy gifts for others? (e.g., birthday, anniversary, etc.)

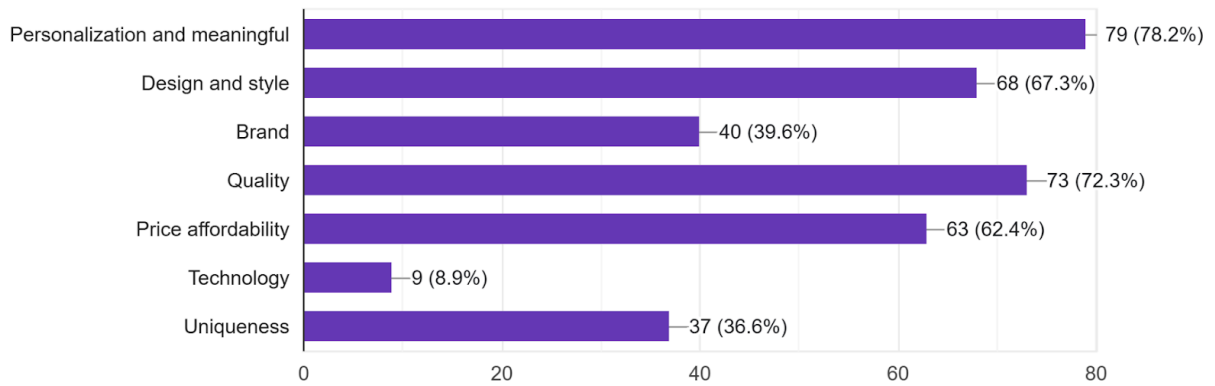
101 responses



Appendix 11: Frequency of Gift Purchases Among Respondents

When buying gifts, what is the most important to you?

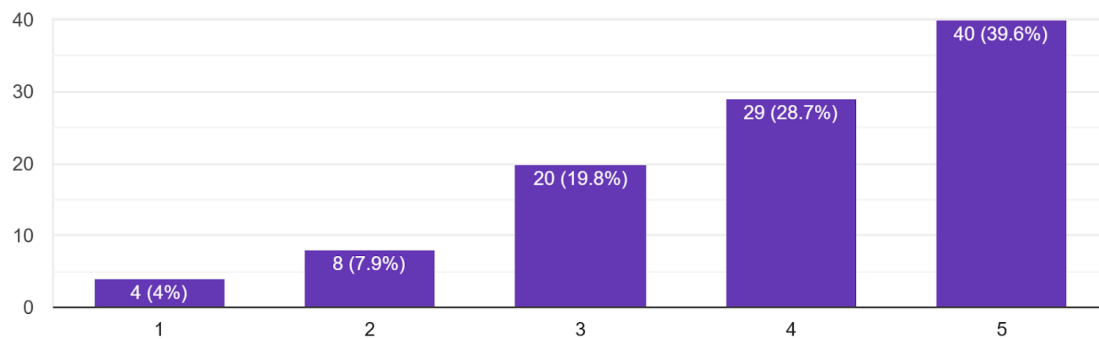
101 responses



Appendix 12: Key Considerations in Gift Selection Among Respondents

How often do you find it hard to choose a meaningful gift?

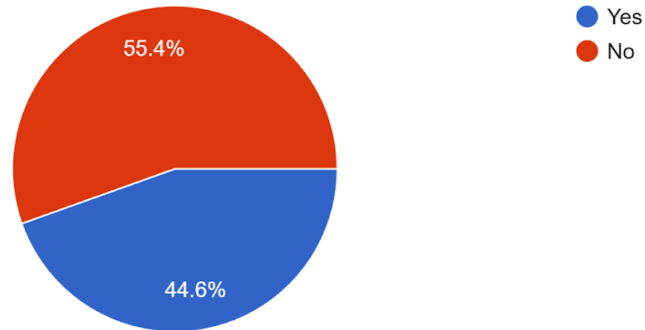
101 responses



Appendix 13: Frequency of Difficulty in Selecting Meaningful Gifts

Have you heard of smart jewelry?

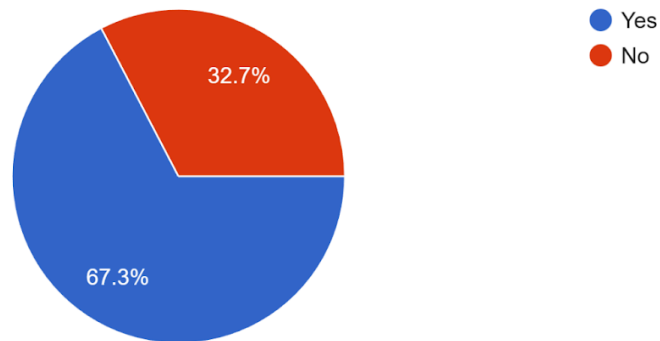
101 responses



Appendix 14: Awareness of Smart Jewelry Among Respondents

Do you wear jewelry?

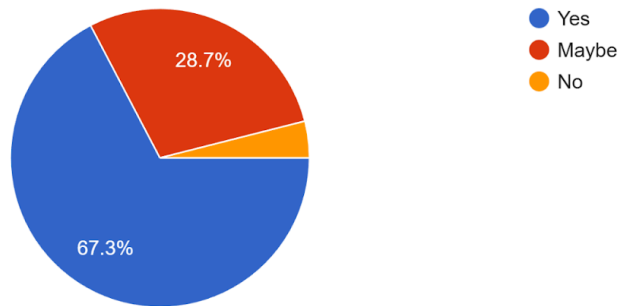
101 responses



Appendix 15: General Jewelry-Wearing Habits Among Respondents

Would you find it interesting to give a piece of jewelry as a gift if it could store a digital message, voice note, photo, video, file, or link that can be un...nce relationship or sharing a special memory, etc.)

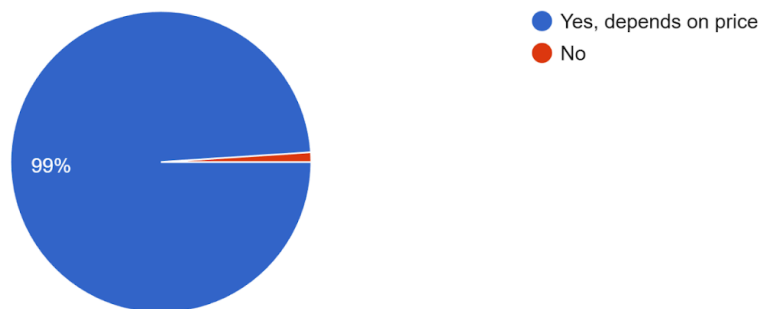
101 responses



Appendix 16: Interest in Giving NFC-Enabled Smart Jewelry as a Gift

Would you be willing to pay more for a jewelry gift that combines style and sentimental technology? (allows the jewelry to store special me...s, or connect with loved ones in a meaningful way)

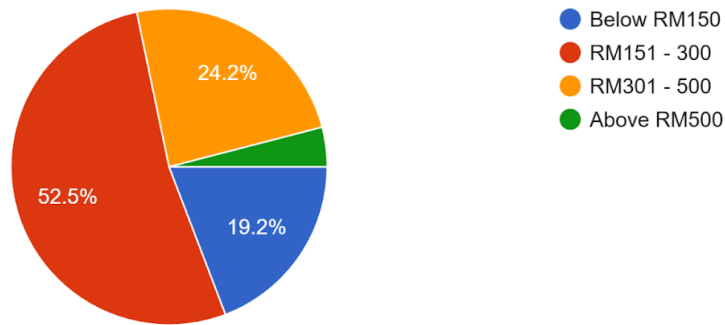
97 responses



Appendix 17: Willingness to Pay More for Jewelry With Sentimental Technology

What price range would you find reasonable for this kind of smart jewelry that includes NFC and blockchain technology?

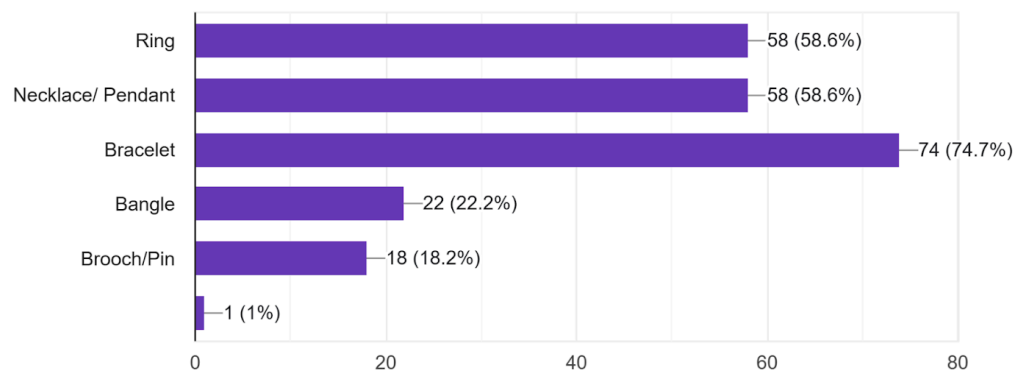
99 responses



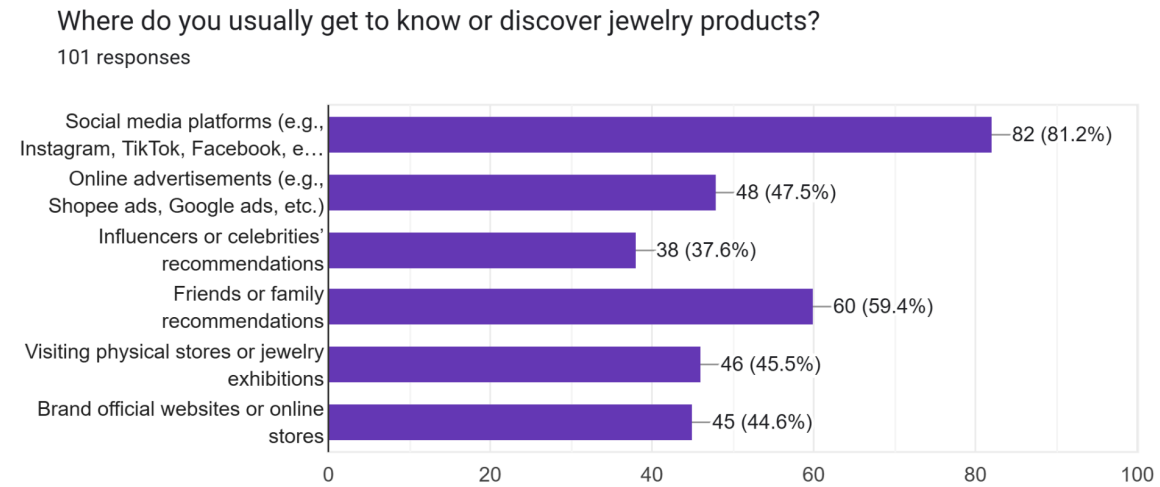
Appendix 18: Reasonable Price Range for NFC & Blockchain-Enabled Smart Jewelry

If you could choose the form of this smart jewelry, which would you prefer?

99 responses



Appendix 19: Preferred Form of Smart Jewelry Among Respondents



Appendix 20: Channels of Discovery for Smart Jewelry Among Respondents

Appendix E

Pandora Case Study

https://drive.google.com/file/d/1ilzUP2KtsxJL3ywljwv_18JgO1S0k55/view?usp=sharing